

RideX

Carpooling that fits your everyday commute

Bengaluru carpooling platform
Ideathon project presentation



Why commuters need a better match

01 Route coordination

Two people may share a road without sharing a usable pickup point.

02 Timing and routine

Daily commutes need compatible departure windows and weekdays.

03 Fair cost sharing

Each passenger should contribute for the distance they share.

04 Trust before the trip

Clear identity status and user-controlled safety tools matter.

The RideX approach

01 Match the shared journey

Rank eligible schedules by route overlap and compatible timing.

02 Support recurring commutes

Save a single trip or plan journeys across selected weekdays.

03 Explain the fare

Split each shared segment among the people onboard.

04 Make safety visible

Show verification status, women-only preferences and SOS tools.

The commuter experience

01 Choose a journey

Select pickup, destination, date, time and flexibility.

02 Find matching schedules

See loading feedback, ranked results and the fare estimate.

03 Save a personal plan

Review a trip or recurring series in My trips.

04 Manage the journey

Cancel plans, enter progress or leave a completed-trip rating.

Current saves do not contact a driver or reserve a real seat.

Bengaluru location coverage

10 real pickup and destination options

Indiranagar Metro Station
Whitefield (Kadugodi) Metro Station
Cubbon Park Metro Station
UB City, Vittal Mallya Road
Pattanagere Metro Station

Majestic Metro Station
Jayanagar Metro Station
Koramangala, 80 Feet Road
Domlur Bus Stand
MG Road Metro Station

Passengers must agree on an exact, safe meeting point.

Matching and distance-based fares

Matching rules

Pickup and drop-off follow the driver's route order.

Weekdays, departure windows and seats must match.

Preferences filter eligible schedules.

Fares use the planning graph. Google road distances do not yet update the estimate.

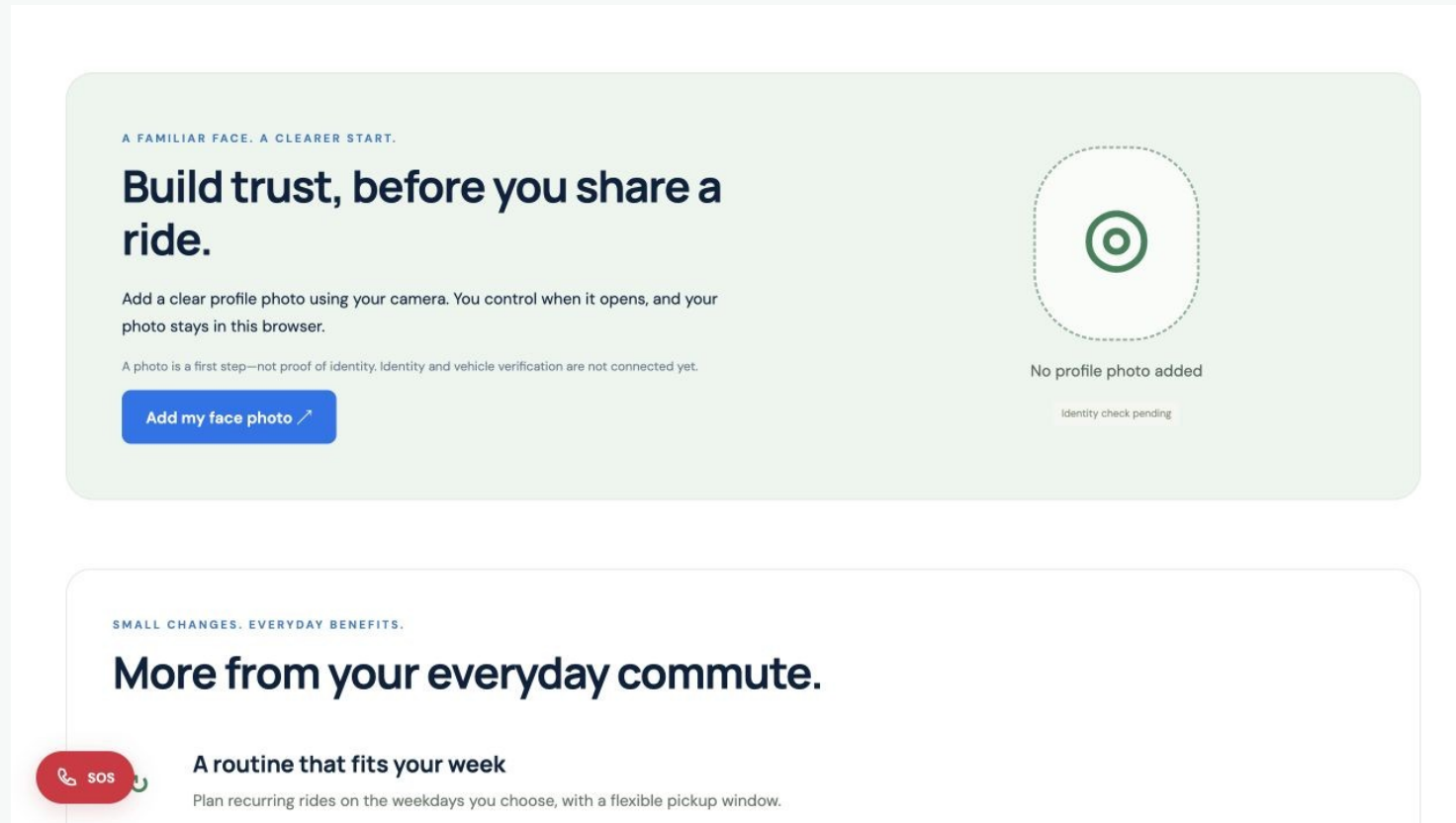
Fare calculation

Each graph segment has a distance and vehicle cost.

The driver and passengers onboard share that segment.

The total uses integer paise to control rounding.

Trust and safety tools



Camera capture

Users choose when to add or remove their profile photo.

Emergency contact

Saved contact, call links and optional location copying.

Independent identity verification remains an integration requirement.

Application architecture

01 Web interface

HTML, CSS and JavaScript for search, plans, profile and safety tools.

02 Journey dashboard

React 19 and Tailwind CSS 4, compiled with Vite 7.

03 Python services

Matching, fare rules, request validation and Google Routes proxy.

04 Planning storage

SQLite locally. Vercel uses isolated requests and a browser-owned snapshot.

Google Maps integration

01 Embedded map

Dark Google Maps styling and the TrafficLayer road overlay.

02 Routes and congestion

Google Routes supplies route geometry, distance, ETA and traffic intervals.

03 Credentials pending

Maps JavaScript and Routes API keys require a billing-enabled Cloud project.

04 Current fallback

An external Google Maps directions link works without embedded credentials.

Driver GPS and traffic-light countdown feeds are not connected.

Validation and current scope

17 Python tests

- Matching and shared fares
- Recurring plans and capacity
- Cancellation, progress and ratings
- Hosted visitor isolation
- Location migration and map parsing

Next integrations

- Authenticated users and shared storage
- Driver acceptance and live GPS
- Independent identity checks
- Payments and emergency dispatch
- Google provider checks with real keys

Existing driver schedules are examples. Hosted plans belong to the visitor's browser.

Team

Six-member team

Tanmay Daga
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Dibakar Das
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Individual contributions await team confirmation.

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Project access and submission

Public GitHub repository

<https://github.com/aaravchoudhary018/FireTech->

Live application

<https://firetech-murex.vercel.app/>

Each member must submit a public repository on their own account.
Confirm whether the event accepts forks before choosing that approach.